

The auto-bot development and rollout were not thoroughly tested for all potential scenarios and, thus, lacked testing rigour, which could have identified problems early on. While the time required to fix the situation was too long, detecting the failure of the auto-bot took considerably longer.

Did you notice a pattern here?

Whether it is chatbot Tay by Microsoft, Uber's autonomous car or the auto-bot of Australian telco, all of these have something in common that failed. And it is not the technology!

In all these scenarios, either the creator of AI or businesses that deployed the AI were not careful enough. They did not follow the fundamental tenet of handling something as powerful as automation or primitive AI responsibly. We often say, "With great power comes great responsibility." And yet, in all these cases, responsible design or deployment did not happen or did not occur in full spirit.

Responsible behaviour is necessary for the deployment and use of AI as well as all other stages from conception to design, testing to implementation, and ongoing management and governance.

Almost all the cases discussed here had a certain level of weakness in the solution conception stage, and this directly seeped into their development.

In cases of a chatbot, telco's auto-bot and autonomous car, emphasis on solution quality was not enough. There might have been a few testing routines—just enough to meet the requirements of IT development frameworks—but not enough to meet the AI development framework, which does not exist!

In the case of Amazon, I would question the skillset of decision-makers. Although it appeared to be an inadvertent decision to roll out same-day deliveries to only a few localities, thoughtful consideration could have

raised these questions in reflection. The same goes for the telco case or lightbulb situation where creators lacked thoughtfulness in the design of the solution.

Nine things to learn from this

While there are several use-cases of AI to learn from, I specifically chose these five, which are indicative of a common issue with AI design, development, deployment and governance—a thoughtful and responsible approach.

In summary, I would say:

1. Data governance is essential from ethical AI's point of view; therefore creators of AI need to ensure they have robust data governance foundations, or their AI applications risk being fed with inappropriate data and breaching several laws.
2. Narrow AI is all about the relation between input and output. You provide input x and get output y, or there is input x to do output y. Either way, the nature of input affects output such that indiscriminate input can lead to adverse outcomes. And this is just one good reason why rigorous testing is so important. We must note that in the case of AI systems, general IT system testing mechanisms are usually not enough.
3. Automated decisions are suitable when there is a large volume of decisions to be made, and the criteria are relatively uniform and uncontested. When discretion and exceptions are required, use automated systems only as a tool to assist humans, or do not use them at all. There are still several applications and use-cases that we cannot define as clearly as a game of chess.
4. Humans must always be kept in the loop no matter what, whether it is during the design phase of AI, or testing or deployment. Majority of AI systems are still kids and need a responsible adult to be in charge. Most importantly, ensuring enough

human resources are available to handle the likely workload is always a good idea.

5. Most importantly, a transparent chain of accountability is necessary. If the answer to the question "who is responsible for the decision made by this AI?" does not yield a single person's name, then this gap needs to be fixed.
6. For customer-facing AI systems, having a good experience at all times is crucial. If customers have a terrible experience, they will lose trust and eventually render the AI solution useless.
7. Fairness, lack of bias, interpretability, explainability, transparency, repeatability and robustness are a few critical aspects that are some of the must-have characteristics for a trustworthy AI solution.
8. As AI systems become more powerful, managing risk is going to be even more critical from good governance and risk management points of view. Having this governance in place is not only an umbrella requirement for the industry but is also a good idea for every business to have inhouse.
9. Ethical aspect must always be upheld, and something should not be done just because it can be done. AI that negatively affects employees, customers or the general public, directly or indirectly, serves no good and should not exist.

American philosopher John Dewey, said, "Failure is not a mere failure. It is instructive. The person who really thinks, learns quite as much from his failures as from his successes."  END

Anand Tamboli

The author is a serial entrepreneur, speaker, award-winning published author and emerging technology thought leader.

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